Year 11 · AQA Level 2 Further Maths · Covered topic

Rearranging formulae and algebraic proof

Revisit and practise · OnMaths · Created by Mr Ali

Today I will learn to...

Goals

1. Rearrange formulae, including when the subject appears twice

2. Prove algebraic statements

Do now: retrieval practice

4 mins

1 Solve $3x + 5 = 20$

$x = 5$

2 Expand $(n + 1)^2$

$n^2 + 2n + 1$

3 Write an expression for any even number

$2n$

4 Make x the subject of $y = x + 7$

$x = y - 7$

Objective 1

Rearrange formulae, including when the subject appears twice

Key idea

Learn

!

Do the same to both sides to get the subject on its own. If the subject appears twice: collect the subject terms on one side, factorise the subject out, then divide.

Objective 1: Rearrange formulae, including when the subject appears twice

I do: example 1

I do

Make r the subject of $A = \pi r^2$

1. Divide by π: $r^2 = A/\pi$
2. Square root both sides
3. $r = \sqrt{(A/\pi)}$

I do: example 2

I do

Make x the subject of $y = (2x + 1)/(x - 3)$

1. Multiply by $(x - 3)$: $y(x - 3) = 2x + 1$
2. $xy - 3y = 2x + 1$
3. $xy - 2x = 3y + 1$
4. $x(y - 2) = 3y + 1$, so $x = (3y + 1)/(y - 2)$

I do: example 3

I do

Make t the subject of a(t + b) = ct

1. Expand: at + ab = ct
2. ab = ct − at
3. ab = t(c − a)
4. t = ab/(c − a)

Mini whiteboards

We do

1. Make h the subject of $V = (1/3)\pi r^2 h$

Answer: $h = 3V/(\pi r^2)$

2. Make x the subject of $ax + b = cx + d$

Answer: $x = (d - b)/(a - c)$

3. Make u the subject of $v = u + at$

Answer: $u = v - at$

4. Make x the subject of $y = \sqrt{(x - 3)}$

Answer: $x = y^2 + 3$

You do: choose your challenge

You do

Bronze

Make x the subject of $y = 5x - 3$

Silver

Make r the subject of $V = (4/3)\pi r^3$

Gold

Make x the subject of $3x + a = bx - 2$

Platinum

Make x the subject of $y = (x + 2)/(x - 5)$

You do: answers

Check

Bronze

$x = (y + 3)/5$

Silver

$r = \sqrt[3]{(3V/(4\pi))}$

Gold

$x = (a + 2)/(b - 3)$

Platinum

$x = (5y + 2)/(y - 1)$

If you finish early

Extra

Bronze

Make t the subject of $v = u + at$

$t = (v - u)/a$

Silver

Make r the subject of $A = \pi r^2 + 5$

$r = \sqrt{((A - 5)/\pi)}$

Gold

Make x the subject of $5x - p = qx + 4$

$x = (p + 4)/(5 - q)$

Platinum

Make y the subject of $x = (y + 1)/(2y - 3)$

$2xy - 3x = y + 1$, so $y = (3x + 1)/(2x - 1)$

Exam question 1

3 marks

Make y the subject of $x = (3y - 2)/(y + 4)$

Space for working

Mark scheme: $x(y + 4) = 3y - 2$ (M1). $xy - 3y = -2 - 4x$ (M1). $y = (4x + 2)/(3 - x)$ (A1)

Exam question 2

3 marks

Make u the subject of $1/f = 1/u + 1/v$

Space for working

Mark scheme: $1/u = 1/f - 1/v$ (M1) $= (v - f)/(fv)$ (M1). $u = fv/(v - f)$ (A1)

Objective 1 achieved

Rearrange formulae, including when the subject appears twice



Objective 2

Prove algebraic statements

✓ Achieved: Rearrange formulae, including when the subject appears twice

Key idea

Learn

!

For any integer use n: even = 2n, odd = 2n + 1, consecutive = n, n + 1. Expand and simplify, then write the result in a form that shows the statement, e.g. 4(…) is a multiple of 4. Finish with a sentence.

Objective 2: Prove algebraic statements

I do: example 1

I do

Prove that the sum of three consecutive integers is a multiple of 3

1. Let the integers be n, n + 1 and n + 2
2. Sum = 3n + 3
3. = 3(n + 1)
4. 3(n + 1) is a multiple of 3

I do: example 2

I do

Prove that the difference between the squares of two consecutive odd numbers is a multiple of 8

1. Odd numbers: $2n + 1$ and $2n + 3$
2. $(2n + 3)^2 - (2n + 1)^2 = 4n^2 + 12n + 9 - 4n^2 - 4n - 1$
3. $= 8n + 8 = 8(n + 1)$
4. $8(n + 1)$ is a multiple of 8

I do: example 3

I do

Prove that $(n + 5)^2 - (n + 3)^2$ is always even

1. $n^2 + 10n + 25 - (n^2 + 6n + 9)$
2. $= 4n + 16$
3. $= 2(2n + 8)$
4. $2 \times$ an integer is even

Mini whiteboards

We do

1. Show that $2(3n + 1) + 4$ is a multiple of 6

Answer: $6n + 6 = 6(n + 1)$

2. Prove that the sum of two odd numbers is even

Answer: $(2a + 1) + (2b + 1) = 2(a + b + 1)$, which is even

3. Show that $3(n + 4) - 3n$ is always 12

Answer: $3n + 12 - 3n = 12$

4. Prove that the product of two odd numbers is odd

Answer: $(2a + 1)(2b + 1) = 2(2ab + a + b) + 1$, which is odd

You do: choose your challenge

You do

Bronze

Prove that the sum of two even numbers is even

Silver

Prove that $(n + 2)^2 - n^2$ is a multiple of 4

Gold

Prove that the sum of the squares of two consecutive integers is odd

Platinum

Prove that $(3n + 1)^2 - (3n - 1)^2$ is a multiple of 12

You do: answers

Check

Bronze

$2a + 2b = 2(a + b)$

Silver

$4n + 4 = 4(n + 1)$

Gold

$n^2 + (n + 1)^2 = 2(n^2 + n) + 1$, which is odd

Platinum

$12n = 12 \times n$

If you finish early

Extra

Bronze

Prove that the sum of an odd number and an even number is odd

$(2a + 1) + 2b = 2(a + b) + 1$

Silver

Prove that $(n + 3)^2 - (n + 1)^2$ is a multiple of 4

$4n + 8 = 4(n + 2)$

Gold

Prove that $n^2 + n$ is always even

$n(n + 1)$: one of two consecutive integers is even

Platinum

Prove that $(2n + 1)^2 + (2n - 1)^2$ is 2 more than a multiple of 8

$8n^2 + 2 = 8 \times n^2 + 2$

Exam question 1

3 marks

Prove that $(2n + 1)^2 - (2n - 1)^2$ is a multiple of 8 for all integers n

Space for working

Mark scheme: $4n^2 + 4n + 1 - (4n^2 - 4n + 1)$ (M1) = 8n (A1). 8n = 8 × n, a multiple of 8 (B1)

Exam question 2

3 marks

Prove that the product of two consecutive even numbers is a multiple of 8

Space for working

Mark scheme: 2n(2n + 2) (M1) = 4n(n + 1) (M1). One of n and n + 1 is even, so n(n + 1) = 2k and the product is 8k (A1)

Objective 2 achieved

Prove algebraic statements



Real life

Apply

The surface area of a cylinder is $A = 2\pi r^2 + 2\pi rh$. Make h the subject.

$h = (A - 2\pi r^2)/(2\pi r)$

Extension: stretch yourself

Stretch

1. Prove that $x^2 - 6x + 10 > 0$ for all values of x

$x^2 - 6x + 10 = (x - 3)^2 + 1$, which is at least 1

2. Make x the subject of $y = \sqrt{((x + 2)/(x - 1))}$

$y^2(x - 1) = x + 2$, so $x(y^2 - 1) = y^2 + 2$ and $x = (y^2 + 2)/(y^2 - 1)$

Exit ticket: exam questions

Exit

1. Make x the subject of $y = 4x + 7$ **(1 mark)**

2. Make x the subject of $xy + 3 = 2x - y$ **(2 marks)**

3. Write $(n + 4)^2 - n^2$ in the form $8(\ldots)$ to prove it is a multiple of 8 **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 $x = (y - 7)/4$ [1]

Q2 $xy - 2x = -y - 3$ (M1). $x = (y + 3)/(2 - y)$ (A1) [2]

Q3 $n^2 + 8n + 16 - n^2$ (M1) $= 8n + 16$ (A1) $= 8(n + 2)$, a multiple of 8 (B1) [3]

Total: ____ / 6

Today I achieved

✔ **Rearrange formulae, including when the subject appears twice**

✔ **Prove algebraic statements**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Make x the subject of $y = (x - 4)/3$ [1]

2. Make r the subject of $A = 4\pi r^2$ [2]

3. Make x the subject of $y = (3x - 1)/(x + 2)$ [3]

4. Prove that the sum of four consecutive integers is always even [2]

5. Prove that $(2n + 3)^2 - (2n - 1)^2$ is a multiple of 8 [3]

Extra practice: Mr Barton Maths: AQA Further Maths → search "Changing the subject"